



JKBOSE 2026

CLASS 11TH

INTRODUCTION TO THREE DIMENSIONAL GEOMETRY

MATHS

LECTURE-2

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TOPICS TO BE COVERED

1

Distance Formula

2

Related Problems

Distance Between Two Points

Suppose that $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ be any two points in the space. Then the distance d b/w A and B is given

by;

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Example: Find the distance b/w points $P(3, 2, 1)$ & $Q(1, 0, 11)$

(Sol) Here by Distance formula;

$$PQ = \sqrt{(1-3)^2 + (0-2)^2 + (11-1)^2}$$

$$\Rightarrow PQ = \sqrt{4 + 4 + 100}$$

$$\Rightarrow PQ = \sqrt{108} = \sqrt{36 \times 3} = 6\sqrt{3} \text{ units.}$$



Remark:



* $PQ + QR = PR$
i.e. Sum of two distances is equal to the third distance, then the points P, Q & R are Collinear.

* $PQ + QR > PR$

$PR + QR > PQ$

$PR + PQ > RQ$

$\Rightarrow P, Q$ & R are non-collinear.



EXAMPLE

Show that the points $P(-2, 3, 5)$, $Q(1, 2, 3)$ and $R(7, 0, -1)$ are collinear.

Proof: By Distance formula, we have

$$PQ = \sqrt{(-2-1)^2 + (3-2)^2 + (5-3)^2} = \sqrt{9+1+4} = \sqrt{14} \text{ units.}$$

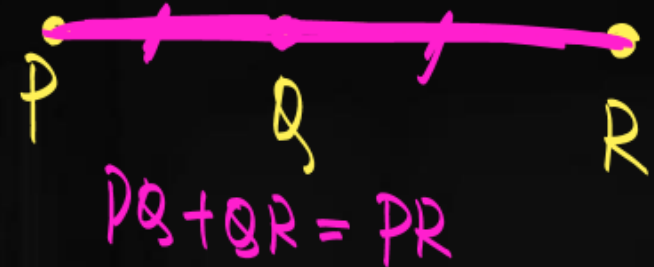
$$QR = \sqrt{(7-1)^2 + (0-2)^2 + (-1-3)^2} = \sqrt{36+4+16} = \sqrt{56} = 2\sqrt{14} \text{ units.}$$

$$PR = \sqrt{(7+2)^2 + (0-3)^2 + (-1-5)^2} = \sqrt{81+9+36} = \sqrt{126} = 3\sqrt{14} \text{ units.}$$

$$\text{So, } PQ + QR = \sqrt{14} + 2\sqrt{14} = 3\sqrt{14}$$

$$\Rightarrow PQ + QR = PR$$

$\Rightarrow P, Q$ and R are collinear points. Hence proved.



EXAMPLE

Find the point on x -axis which is equidistant from the point $A(3, 2, 2)$ and $B(5, 5, 4)$.

(Sol) Suppose that $C(a, 0, 0)$ be the given point
Which is equidistant from $A(3, 2, 2)$ and $B(5, 5, 4)$

$$\text{i.e. } AC = BC$$

$$\Rightarrow AC^2 = BC^2$$

$$\Rightarrow (a-3)^2 + 2^2 + 2^2 = (a-5)^2 + 5^2 + 4^2$$

$$\Rightarrow a^2 + 9 - 6a + 8 = a^2 + 25 - 10a + 41$$

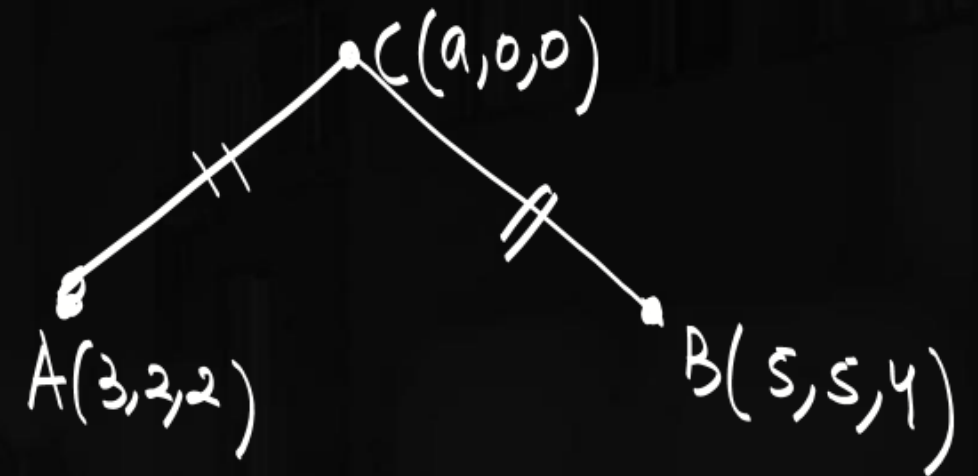
$$\Rightarrow -6a + 10a = 66 - 17$$

$$\Rightarrow 4a = 49$$

$$\Rightarrow a = \frac{49}{4}$$

So, the required point
is $\left(\frac{49}{4}, 0, 0\right)$

Hence the result.



QUESTION (NCERT EXAMPLER):

Distance of the point $(3, 4, 5)$ from the origin $(0, 0, 0)$ is

1 $\sqrt{50}$

2 $\sqrt{51}$

3 7

4 $\sqrt{52}$

$$d = \sqrt{(3-0)^2 + (4-0)^2 + (5-0)^2}$$

$$d = \sqrt{9 + 16 + 25}$$

$$\Rightarrow d = \sqrt{50} \text{ units.}$$

QUESTION

The point on Y-axis which is at a distance of $\sqrt{10}$ units from the point $(1, 2, 3)$ is :

- 1 $(0, 3, 0)$
- 2 $(0, 2, 0)$
- 3 $(0, 1, 0)$
- 4 None of the above

(Sol) Suppose that $P(0, a, 0)$ be the required point which is at distance of $\sqrt{10}$ units from $Q(1, 2, 3)$

$$\text{i.e. } PQ = \sqrt{10}$$

$$\Rightarrow \sqrt{1^2 + (2-a)^2 + 3^2} = \sqrt{10}$$

$$\Rightarrow 1 + (2-a)^2 + 9 = 10$$

$$\Rightarrow 10 + (2-a)^2 = 10$$

$$\Rightarrow (2-a)^2 = 0$$

$$\Rightarrow 2-a = 0$$

$$\Rightarrow a = 2$$

So, the required point is $P(0, 2, 0)$
Hence the result.

QUESTION (NCERT EXAMPLER):

If the distance between the points $(a, 0, 1)$ and $(0, 1, 2)$ is $\sqrt{27}$, then the value of a is.

- 1 5 ✗
- 2 ± 5 ✓
- 3 -5 ✗
- 4 None of these

(Sol) let the given points be $P(a, 0, 1)$ & $Q(0, 1, 2)$

Such that $PQ = \sqrt{27}$ units.

$$\Rightarrow \sqrt{a^2 + 1^2 + 1^2} = \sqrt{27}$$

$$\Rightarrow \sqrt{a^2 + 2} = \sqrt{27}$$

$$\Rightarrow a^2 + 2 = 27$$

$$\Rightarrow a^2 = 25$$

$$\Rightarrow a = \pm 5$$

$$\Rightarrow a = \pm 5$$

QUESTION

The equation of the set of points which are equidistant from the points $(1, 2, 3)$ and $(3, 2, -1)$ is :

1 $x - 3z = 0$

2 $x - 2z = 0$

3 $x - 4z = 0$

4 $x + 2z = 0$

(Sol) Suppose that $R(x, y, z)$ be the point which is equidistant from $P(1, 2, 3)$ and $Q(3, 2, -1)$.

i.e. $PR = QR$

$$\Rightarrow \sqrt{(x-1)^2 + (y-2)^2 + (z-3)^2} = \sqrt{(x-3)^2 + (y-2)^2 + (z+1)^2}$$

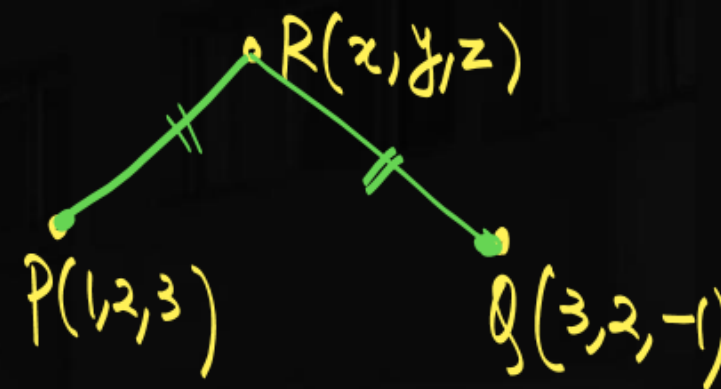
$$\Rightarrow (x-1)^2 + (y-2)^2 + (z-3)^2 = (x-3)^2 + (y-2)^2 + (z+1)^2$$

$$\Rightarrow x^2 - 2x + 1 + z^2 - 6z + 9 = x^2 - 6x + 9 + z^2 + 2z + 1$$

$$\Rightarrow -2x - 6z + 10 = -6x + 2z + 10$$

$$\Rightarrow -2x + 6x - 6z - 2z = 0$$

$$\Rightarrow 4x - 8z = 0$$



$$\Rightarrow 4(x - 2z) = 0$$

$$\Rightarrow x - 2z = 0$$

QUESTION

In $\triangle ABC$, the mid points of the sides AB, BC and CA are $(l, 0, 0)$, $(0, m, 0)$ and $(0, 0, n)$ respectively. Then $\frac{AB^2 + BC^2 + CA^2}{l^2 + m^2 + n^2}$ is equal to :

1

2 ~~X~~

(Sol) Here $\frac{x_1 + x_2}{2} = l, \frac{y_1 + y_2}{2} = 0, \frac{z_1 + z_2}{2} = 0$

2

4 ~~X~~

$$\frac{x_2 + x_3}{2} = 0, \frac{y_2 + y_3}{2} = m, \frac{z_2 + z_3}{2} = 0$$

3

8 ✓

$$\frac{x_1 + x_3}{2} = 0, \frac{y_1 + y_3}{2} = 0, \frac{z_1 + z_3}{2} = n$$

4

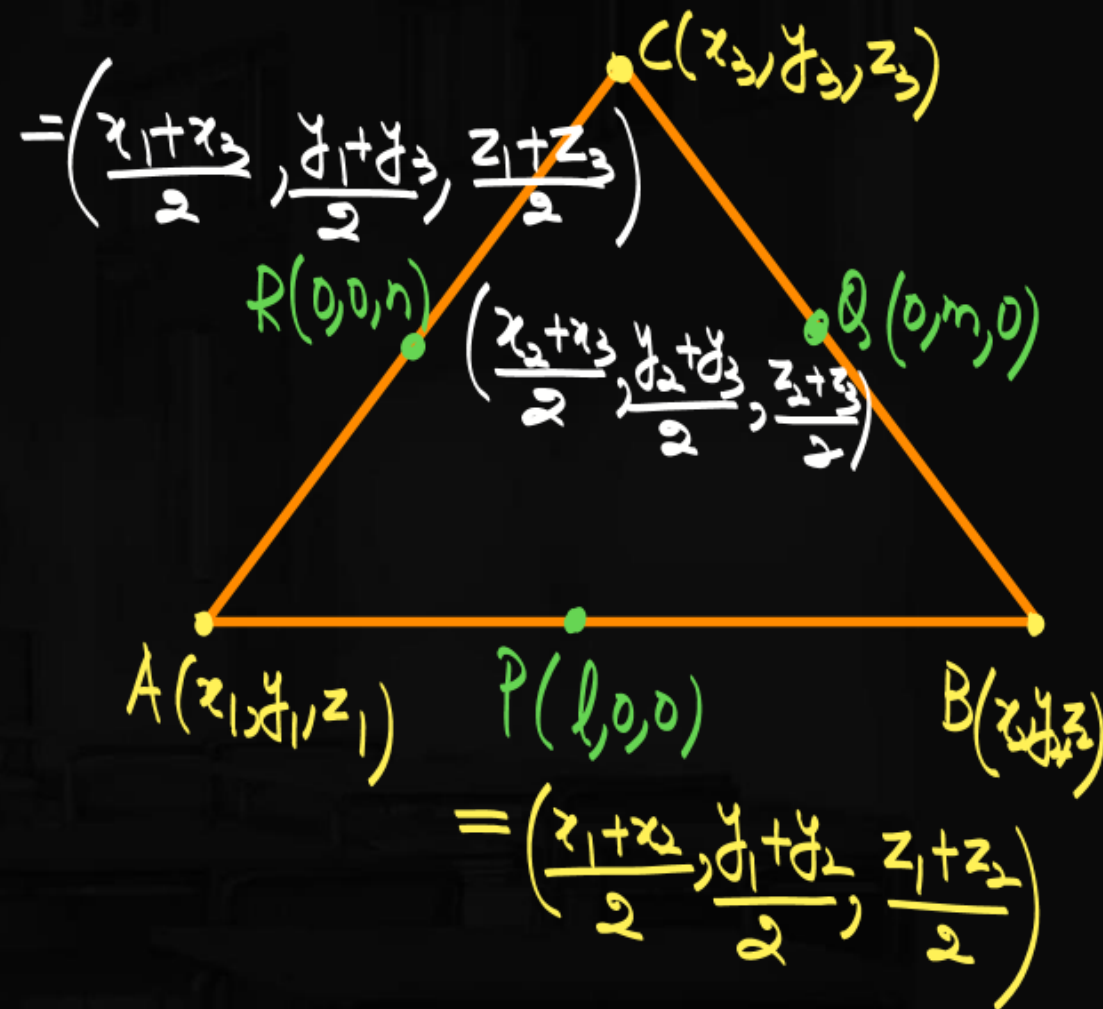
16 ~~X~~

Consider the following:

$$x_1 + x_2 = 2l \quad \text{--- (1)}$$

$$x_2 + x_3 = 0 \Rightarrow x_3 = -x_2 \quad \text{--- (2)}$$

$$x_1 + x_3 = 0 \Rightarrow x_1 - x_2 = 0 \quad \text{--- (3)}$$



Adding ① & ③, we get;

$$x_1 + \cancel{x_2} + x_1 - \cancel{x_2} = 2l$$

$$\Rightarrow 2x_1 = 2l$$

$$\Rightarrow x_1 = l$$

$$\text{So, } x_2 = l, x_3 = -l$$

Similarly, we can show that

$$y_1 = -m, y_2 = m, y_3 = m$$

$$z_1 = n, z_2 = -n, z_3 = n$$

$$\text{Therefore } AB^2 + BC^2 + CA^2 = (2m)^2 + (-2n)^2 + (2l)^2 + (2n)^2 + (-2l)^2 + (2m)^2$$

$$\Rightarrow (AB)^2 + (BC)^2 + (CA)^2 = 4m^2 + 4n^2 + 4l^2 + 4n^2 + 4l^2 + 4m^2$$

$$\Rightarrow (AB)^2 + (BC)^2 + (CA)^2 = 8m^2 + 8l^2 + 8n^2$$

$$\Rightarrow \frac{(AB)^2 + (BC)^2 + (CA)^2}{l^2 + m^2 + n^2} = \frac{8(l^2 + m^2 + n^2)}{(l^2 + m^2 + n^2)}$$

$$\Rightarrow \frac{(AB)^2 + (BC)^2 + (CA)^2}{l^2 + m^2 + n^2} = 8$$

Tawkeer Amin Sofi



Rasib Ayoub Rattex



Syed Fakhir



THANK

You